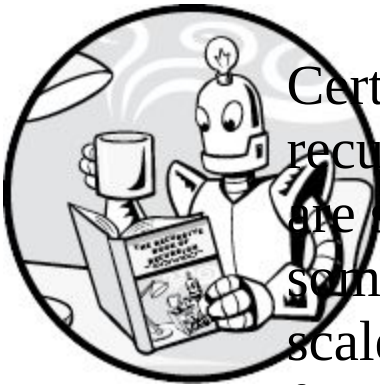


9

DRAWING FRACTALS



Certainly, the most fun application of recursion is drawing fractals. *Fractals* are shapes that repeat themselves, sometimes chaotically, at different scales. The term was coined by the founder of fractal geometry, Benoit B. Mandelbrot, in 1975 and is derived from the Latin *frāctus*, meaning broken or fractured, like shattered glass. Fractals include many natural and artificial shapes. In nature, you might see them in the shapes of trees, fern leaves, mountain ranges, lightning bolts, coastlines, river networks, and snowflakes. Mathematicians, programmers, and artists can create elaborate geometric shapes based on a few recursive rules.

Recursion can produce elaborate fractal art using surprisingly few lines of code. This chapter covers Python's built-in `turtle` module for generating several common fractals with code. To create turtle graphics with JavaScript, you can use Greg Reimer's `jtg` library. For simplicity, this chapter presents only the Python fractal drawing programs and not the